

2022 | ④ last part:

$$N \equiv n_x^2 + n_y^2 + n_z^2$$

$$E_N = \frac{\hbar^2 \pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)$$

E_N : deg 3:

1

$$E_1 \begin{cases} \textcircled{a} n_x = 1, & n_y = n_z = 0 \\ \textcircled{b} n_y = 1, & n_x = n_z = 0 \\ \textcircled{c} n_z = 1, & n_x = n_y = 0 \end{cases} \quad \text{deg } 3$$

$$E_2: 2 = n_x^2 + n_y^2 + n_z^2$$

$$\begin{cases} \textcircled{a} n_x = 1, & n_y = 1, & n_z = 0 \\ \textcircled{b} n_x = 1, & n_z = 1, & n_y = 0 \\ \textcircled{c} n_y = 1, & n_z = 1, & n_x = 0 \end{cases} \quad \text{deg } 3$$

ex: $n_x = 2$

2^2 : cut do

$$E_3: n_x = 1, n_y = 1, n_z = 1 \quad \text{deg } 1$$

$$E_4: \begin{cases} n_x = 2, & n_y = 0, & n_z = 0 \\ n_y = 2, & n_x = n_z = 0 \\ n_z = 2, & n_x = n_y = 0 \end{cases} \quad \text{deg } 3$$

⑩ 2022: $\psi(x) = \begin{cases} A\sqrt{x(1-x)} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} \textcircled{a} \bullet \int_{1/2}^1 |\psi|^2 dx &= \int_{1/2}^1 A^2 x(1-x) dx = \\ &= |A|^2 \int_{1/2}^1 (x - x^2) dx = |A|^2 \left(\frac{1}{2} x^2 \Big|_{1/2}^1 - \frac{1}{3} x^3 \Big|_{1/2}^1 \right) \end{aligned}$$

$$|A|^2 \left(\frac{1}{2} \left(1 - \left(\frac{1}{2} \right)^2 \right) - \frac{1}{3} \left(1 - \left(\frac{1}{2} \right)^3 \right) \right)$$

$$\hookrightarrow |A|^2 \int_0^1 (x - x^2) dx = 1$$

$$\begin{aligned}
 &= |A|^2 \left(\frac{1}{2} x^2 \Big|_0^1 - \frac{1}{3} x^3 \Big|_0^1 \right) \\
 &= |A|^2 \left(\frac{1}{2} - \frac{1}{3} \right) = 1 \\
 &\quad |A|^2 = 6
 \end{aligned}$$

$$\begin{aligned}
 P(x \geq 1/2) &= 6 \left(\frac{1}{2} \left(1 - \left(\frac{1}{2} \right)^2 \right) - \frac{1}{3} \left(1 - \left(\frac{1}{2} \right)^3 \right) \right) \\
 &= \frac{42}{72} \approx 0.58
 \end{aligned}$$

(a) Polar / spherical: $\psi(r, \theta, \phi)$

$$* \boxed{P(x \in [a, b]) = \int \int \int \underbrace{r^2 \sin(\theta)}_{\substack{r \in [0, 1] \\ \theta \in [0, \pi] \\ \phi \in [0, 2\pi]}} |\psi|^2 dr d\phi d\theta} *$$

$$\begin{cases} r \in [0, 1] \\ \theta \in [0, \pi] \\ \phi \in [0, 2\pi] \end{cases}$$

$$P(r \geq 1/2) = \underbrace{\left(\int_0^\pi \sin(\theta) d\theta \right)}_2 \underbrace{\left(\int_0^{2\pi} d\phi \right)}_{2\pi} \underbrace{\left(\int_{1/2}^1 r^2 |B|^2 (r - r^2) dr \right)}_{\text{yellow arrow}}$$

$$\int_0^\pi \sin(\theta) d\theta = -\cos(\theta) \Big|_0^\pi = -(-1 - 1) = 2$$

$$P(r \geq 1/2) = 4\pi |B|^2 \int_{1/2}^1 (r^3 - r^4) dr$$

$$\begin{aligned}
 &= 4\pi |B|^2 \left(\frac{1}{4} r^4 - \frac{1}{5} r^5 \right) \Big|_{1/2}^1 * \\
 &= 4\pi |B|^2 \left(\frac{1}{4} \left(1 - \left(\frac{1}{2} \right)^4 \right) - \frac{1}{5} \left(1 - \left(\frac{1}{2} \right)^5 \right) \right)
 \end{aligned}$$

$$P_{\text{tot}} = 1 = \underbrace{\int_0^\pi \sin(\theta) d\theta}_2 \underbrace{\int_0^{2\pi} d\phi}_{2\pi} \int_0^1 |B|^2 (r^3 - r^4) dr$$

$$1 = 4\pi |B|^2 \int_0^1 (r^3 - r^4) dr$$

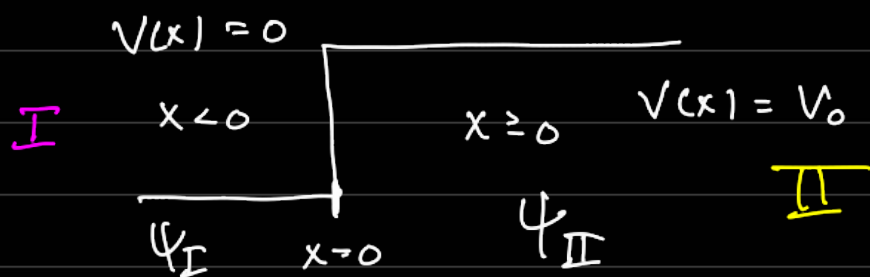
$$\left(\frac{1}{4} - \frac{1}{5} \right)$$

$$\frac{1}{20}$$

$$1 = \frac{4\pi |B|^2}{20} \Rightarrow 4\pi |B|^2 = 20$$

$$P(r > 1/2) = 20 \left(\frac{1}{4} \left(1 - \left(\frac{1}{2} \right)^4 \right) - \frac{1}{5} \left(1 - \left(\frac{1}{2} \right)^5 \right) \right)$$

2023 | Q3



Particle has $E > V_0$

$$\text{I) } \psi_I(x) = e^{ikx} + \sqrt{R} e^{-ikx}$$

$$\text{II) } \psi_{II}(x) = \sqrt{T \left(\frac{k}{k'} \right)} e^{ik'x}$$

$$\text{ii) } \hat{H}|\psi\rangle = E|\psi\rangle$$

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

$$\text{(I) } \hat{H}_I = \frac{p^2}{2m}$$

$$\text{(II) } \hat{H}_{II} = \frac{p^2}{2m} + V_0$$

$$\hat{H}_I = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$$

$$\hat{H}_{II} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V_0$$

$$\text{(I) } -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E \psi$$

SHM:

$$\left[\frac{\partial^2 \psi}{\partial x^2} = - \underbrace{\left(\frac{2mE}{\hbar^2} \right)}_{k^2} \right] \psi$$

$$\frac{\partial^2 f}{\partial t^2} = -\omega^2 f$$

$$f \sim e^{\pm i\omega t}$$

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2 \psi \quad \text{*}$$

$$\text{sol}^n : \psi_I(x) = e^{ikx} + \sqrt{R} e^{-ikx}$$

$$k^2 = \left[\frac{2mE}{\hbar^2} \right]$$

$$\textcircled{\text{II}} \quad \hat{H}_{II} |4\rangle = E |4\rangle$$

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi \right) = E \psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = (E - V_0) \psi$$

$$\frac{\partial^2 \psi}{\partial x^2} = - \underbrace{\left[\frac{2m}{\hbar^2} (E - V_0) \right]}_{(k')^2} \psi$$

$$E > V_0$$

$$E - V_0 > 0$$

$$(k')^2 = \frac{2m}{\hbar^2} (E - V_0)$$

(ii) look for $T \in \mathbb{R}$:

$$\textcircled{a} \quad \text{B.C. 1: } \psi_I(0) = \psi_{II}(0) \quad *$$

$$1 + \sqrt{R} = \sqrt{T \left(\frac{k'}{k} \right)}$$

(a)

$$\textcircled{b} \quad \text{B.C. 2: } \psi'_I(0) = \psi'_{II}(0) \quad *$$

$$ik + \sqrt{R}(-ik) = \sqrt{T \left(\frac{k'}{k} \right)} \cdot (ik')$$

(b)

(α): Solve for $T(\frac{k}{k'})$ and plug into (β)

$$ik + \sqrt{R}(-ik) = (1 + \sqrt{R})(ik')$$

solve for $\sqrt{R}$

$$k + \sqrt{R}(-k) = k' + k'\sqrt{R}$$

$$\sqrt{R}(k' + k) = k - k'$$

$$\boxed{\sqrt{R} = \frac{k - k'}{k' + k}} \quad \leftarrow \text{plug in } k, k' \text{ from above}$$

now, plug back into (α) to find T

$$1 + \left(\frac{k - k'}{k' + k} \right) = \sqrt{T\left(\frac{k}{k'}\right)}$$

Solve for T :

